

Armaan Dhah

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EDUCATION

University of California, Los Angeles | Los Angeles, CA

Sep. 2025 – June 2027

B.S. Mechanical Engineering; GPA: 3.7/4.0

Clovis Community College | Fresno, CA

Aug. 2023 – May 2025

Mechanical Engineering; GPA: 4.0/4.0

SKILLS

Manufacturing: 3D Printing, Lathe, Mill, CNC Mill, Soldering, TIG Welding, Spot Welding, Drill Press, Angle Grinder, Bandsaw

Software: SolidWorks, Finite Element Analysis (Ansys and SolidWorks), FusionCAM, Altium Co-Designer, MATLAB, AutoCAD, Microsoft Suite

Certifications: Certified SolidWorks Associate (CSWA)

WORK EXPERIENCE

Advanced Spade Company

May 2026 – Present

Mechanical Engineering Intern

- Redesigned a pitch/roll camera gimbal for a sewer-surveying robot, cutting package volume 53% through re-tuned gear reduction, a ball-bearing-supported roll shaft, and slip-ring integration, programming rotary encoders with a Teensy 4.1 and Arduino IDE.
- Led mechanical design across three phases of an in-house GNSS positioning cell device to replace third-party units, building test fixtures to validate candidate chips, packaging a 4-PCB production device around a parametric housing, and designing and assembling the custom 2s2p battery pack.
- Designed a single-slot external battery charger enclosure in SolidWorks, converging on a slip-fit retention geometry through 13 design-table iterations.
- Designed and fabricated a T-slot aluminum Testbench Mobility Unit to convert the CRKT CNC testbench into a mobile unit.
- Deployed an automated 3D-printing farm to streamline prototyping and production of 3D-printed components across the company's product line.

Clovis Community College Tutorial Center | Fresno, CA

Aug. 2024 – May 2025

Team Lead – Student Aide II

- Provided math, physics, chemistry, and engineering tutoring, supporting students in developing problem-solving skills and conceptual understanding.

ENGINEERING PROJECTS

UCLA Bruin Baja SAE | Los Angeles, CA

Suspension Lead Engineer

May 2026 – Present

- Direct a team of 13 project engineers through the design, analysis, fabrication, and validation of the entire suspension and steering system, coordinating directly with other subsystem leads to ensure flawless vehicle integration.
- Optimize suspension geometry to elevate vehicle handling and reliability while supervising component packaging, mass reduction, and strict rule compliance.
- Manage the subteam budget and defend design architectures at formal technical reviews and competition design presentations.

Suspension Engineer

Dec. 2025 – May 2026

- Fabricated suspension components for upper and lower A-arms by generating 1:1 DXF templates for accurate tube notching and machining parts such as high-misalignment spacers on manual mill and lathe; assisted in CNC milling tie rods and final suspension assembly.

General Member

Sept. 2025 – Dec. 2025

- Designed and fabricated a scaled mock centerlock assembly in SolidWorks for integration into a software testbench, enabling software and electrical teams to validate control systems independent of vehicle readiness.
- Implemented an axially repositionable limit switch assembly to optimize switch placement during testing and inform improvements for the next vehicle design cycle.

Flagship Combat Robotics - UCLA American Society of Mechanical Engineers | Los Angeles, CA

Sept. 2025 – May 2026

SKULL CRUSHER Project Engineer

- Designed and manufactured a drisk-style spinning weapon system using SolidWorks for the 30 lb combat robot Skull Crusher, iterating geometry and packaging to maintain strength and functionality while meeting a 12 lb weapon mass allocation.
- Performed ANSYS FEA across multiple design revisions to predict stress concentrations and loading responses, validating spline hub survivability under impact forces expected in combat.
- Machined critical components using manual lathe and mill operations, and programmed toolpaths in Fusion CAM to manufacture complex parts on a CNC mill.

Crush Depth MATE ROV Underwater Robotics | Fresno, CA

Aug. 2024 – June 2025

FLOAT Lead Mechanical Engineer

- Led a sub-team in the end-to-end design and fabrication of an underwater vertical profiling float, managing mechanical design, buoyancy control, and waterproofing integration.
- Modeled eight iterations of buoyancy engine and housing assemblies in SolidWorks CAD; 3D printed components and applied custom acetone vapor finishing to improve tensile strength, surface quality, and sealing performance.
- Engineered a precision O-ring waterproofing interface with optimized 25% compression geometry; depth-tested over 150 hours to achieve 3.0 meters with zero leakage, representing a 90% improvement over previous years.
- Supervised teammates in CAD modeling, fabrication, and mechanical testing, ensuring milestone completion under tight competition deadlines; submitted the system to the 2025 MATE ROV Explorer Class Championship.